on. Or, think of an organ pipe or a French horn. The sounds that each of these instruments can produce are a direct consequence of the resonant patterns of vibrating air streams in their interior; these are determined by the precise size and shape of the spatial surroundings within the instrument through which the air streams are channeled. Curled-up spatial dimensions have a similar impact on the possible vibrational patterns of a string. Since tiny strings vibrate through all of the spatial dimensions, the precise way in which the extra dimensions are twisted up and curled back on each other strongly influences and tightly constrains the possible resonant vibrational patterns. These patterns, largely determined by the extradimensional geometry, constitute the array of possible particle properties observed in the familiar extended dimensions. This means that extradimensional geometry determines fundamental physical attributes like particle masses and charges that we ovserve in the usual three large space dimensions of common experience.

This is such a deep and important point that we say it once again, with feeling. According to string theory, the universe is made up of tiny strings whose resonant patterns of vibration are the microscopic origin of particle masses and force charges. String theory also requires extra space dimensions that must be curled up to a very small size to be consistent with our never having seen them. But a tiny string can probe a tiny space. As a string moves about, oscillating as it travels, the geometrical form of the extra dimensions plays a critical role in determining resonant patterns of vibration. Because the patterns of string vibrations appear to us as the masses and charges of the elementary particles, we conclude that these fundamental properties of the universe are determined, in large measure, by the geometrical size and shape of the extra dimensions. That’s one of the most far-reaching insights of string theory.

Since the extra dimensions so profoundly influence basic physical properties of the universe, we should now seek—with unbridled vigor—an understanding of what these curled-up dimensions look like.

What Do the Curled-Up Dimensions Look Like?

The extra spatial dimensions of string theory cannot be “crumpled” up any which way; the equations that emerge from the theory severely restrict the geometrical form that they can take. In 1984, Philip